

# Direction Instability Identifies Drug Mechanism Transport While Geometric Alternatives Measure Magnitude

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## Abstract

High-throughput perturbation screens produce multi-dimensional signatures for thousands of drugs and gene knockdowns across panels of cell lines. Geometric metrics—direction instability, Grassmannian geodesic distance, profile variability, optimal transport distances—are used to quantify whether a perturbation acts consistently across cellular contexts. We show that three of these four metric families are dominated by estimation quality: any variable that improves the precision of the estimated object (effect magnitude, sample size) mechanically shifts the distance metric. Across three perturbation modalities (LINCS L1000, Perturb-seq, DepMap CRISPR screens) and two measurement technologies (transcriptomic and morphological), the same artifact produces effects ranging from large (DepMap Cohen’s  $d = 2.65$ , Perturb-seq geodesic  $d = 0.55$ ) to ambiguous (cross-cell-type transport  $d = -0.14$  to  $-0.172$ ), all of which invert or vanish after magnitude control. The fourth metric family, cosine-based direction instability, is immune by construction: it normalizes by magnitude before comparison. This immunity is functional: on a pre-registered construction-neutral benchmark, direction instability predicts cross-cell-line mechanism transport (leave-one-out AUROC = 0.945, 95% CI [0.937, 0.952],  $n = 2,694$  LINCS drugs), outperforming all alternatives including Fréchet variance (0.857, [0.841, 0.872]). On the cosine-matched outcome, Fréchet variance is marginally higher (0.976 vs. 0.969,  $\Delta = -0.007$ ,  $[-0.012, -0.002]$ )—consistent with the construction-matching critique this paper advances. Direction instability stratifies drugs by mechanism of action (machinery vs. receptor  $d = 1.94$  in LINCS,  $d = 0.77$  in JUMP Cell Painting) and reveals an HDAC inhibitor selectivity gradient visible only through directional analysis. Three pre-registered experiments (Perturb-seq cosine transport, construction-neutral LINCS benchmark, cross-modality MOA stratification) yielded five of ten criteria passing and five failing, with each failure interpretable. Pre-registered criteria and analysis code were frozen before experiments (commits 1dc20a2, 249abaf, 8d74bd0). The corrective rule: normalize for effect magnitude before comparing groups defined by effect strength, or geometric “divergence” reduces to the tautology that bigger effects are noisier.

## 1 Introduction

High-throughput perturbation screens produce multi-dimensional signatures for thousands of drugs and gene knockdowns across panels of cell lines. A natural question follows: does this perturbation act consistently across contexts, or does it produce divergent effects depending on cellular background? Geometric metrics formalize this question. Direction instability measures the mean pairwise angular deviation of perturbation signatures across cell lines. Grassmannian geodesic distance measures how much a perturbation’s response subspace rotates between cell types. Profile variability (standard deviation across contexts) measures raw spread.

These metrics are used to characterize drug mechanism conservation, gene essentiality, and perturbation quality in LINCS L1000 [Subramanian et al., 2017], Perturb-seq [Replogle et al., 2022], JUMP Cell Painting [Chandrasekaran et al., 2023], and DepMap CRISPR screens. The implicit assumption is that “more divergent” reflects biology—context-dependent mechanisms, compensatory responses, or pathway rewiring.

This paper makes two contributions. First, we demonstrate that variance-based, subspace-based, and distribution-based formulations of these metrics are dominated by estimation quality—any variable that improves the precision of the estimated object (effect magnitude, sample size) mechanically shifts the distance metric. Cosine-based metrics (direction instability) are immune because they normalize by magnitude before comparison. Second, we exploit this immunity to show that direction instability captures genuine biological signal: it predicts cross-cell-line mechanism transport in LINCS L1000 (leave-one-out AUROC = 0.945, 95% CI [0.937, 0.952] on a construction-neutral outcome), stratifies drugs by mechanism of action across two measurement technologies, and reveals an HDAC inhibitor selectivity gradient that variance-based metrics cannot isolate.

Our evidence comes from a sequence of pre-registered and exploratory experiments across three data modalities, reported with an explicit integrity boundary between confirmatory and post-hoc analyses. The discovery unifies two previously puzzling observations: (1) biological confound corrections fail to rerank direction instability across three domains (Spearman  $\rho = 0.83$ – $0.99$ ), and (2) essential gene knockdowns appear to “diverge” more than non-essentials ( $d = 0.55$  in Perturb-seq). The first reflects genuine directional stability of perturbation signatures. The second reflects magnitude-confounding of the comparison. Both dissolve under the same DAG.

## 2 Data

### 2.1 LINCS L1000

We used Level 5 z-score profiles from LINCS L1000 [Subramanian et al., 2017] (GEO accession GSE92742), covering 978 landmark genes across up to 70 cell lines. After filtering to drugs tested in  $\geq 3$  cell lines, the dataset contains 8,949 compounds. Drug annotations (mechanism of action, target, clinical phase) were obtained from the Drug Repurposing Hub [Corsello et al., 2017]. For the benchmark analysis (Section 7.3), the threshold was raised to  $\geq 10$  cell lines, yielding 2,694 drugs.

### 2.2 Perturb-seq

We used genome-scale Perturb-seq data from Replogle et al. [2022], covering 1,676 gene knockdowns in K562 and RPE1 cell types. Pseudobulk signatures were computed as mean expression across cells sharing the same guide, restricted to genes with  $\geq 20$  cells in both cell types. Subspace data (top-5 PCA subspaces per gene per cell type) were extracted from the same dataset for geodesic distance and EMD analyses.

### 2.3 DepMap CRISPR screens

We used Chronos dependency scores from the DepMap 24Q4 release (1,178 cell lines). Essential genes were defined by DepMap’s common-essentials list ( $n = 1,242$ ); non-essential genes by the nonessentials list ( $n = 726$ ). PC1 was removed before computing variability to control for global fitness effects (PC1 removal changed the effect by 0.2%).

## 2.4 JUMP Cell Painting

We used morphological profiles from the JUMP Cell Painting dataset [Chandrasekaran et al., 2023], covering 25,254 compounds across 3,180 features in U2OS cells. Compounds were matched to LINCS drugs by InChIKey prefix (first 14 characters), yielding 330 matched compounds with MOA annotations.

## 3 Metrics and Their Magnitude Sensitivity

### 3.1 Direction instability (cosine-based, magnitude-invariant)

For a perturbation tested in  $K$  contexts with signature vectors  $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^d$ :

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \frac{\mathbf{s}_i^\top \mathbf{s}_j}{\|\mathbf{s}_i\| \|\mathbf{s}_j\|} \quad (1)$$

The  $\ell_2$  normalization makes  $D$  invariant to the magnitude of each  $\mathbf{s}_i$ . A drug that produces a strong, consistent signal across cell lines receives the same  $D$  as one producing a weak, consistent signal.

### 3.2 Profile variability (variance-based, magnitude-sensitive)

For a gene with dependency scores  $x_1, \dots, x_K$  across  $K$  cell lines:

$$V = \text{SD}(x_1, \dots, x_K) \quad (2)$$

This metric grows with the absolute magnitude of effects. A gene with scores uniformly near  $-1$  (strongly essential everywhere) and one with scores clustered near 0 (non-essential everywhere) will have different  $V$  even if both are equally “consistent”—the essential gene’s scores simply occupy a larger range.

### 3.3 Grassmannian geodesic distance (subspace-based, indirectly magnitude-sensitive)

For a gene knocked down in two cell types, let  $U_1, U_2 \in \text{Gr}(k, d)$  denote the top- $k$  PCA subspaces of the perturbation response. The geodesic distance  $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$  (principal angles) is geometrically magnitude-invariant. However,  $U_1$  and  $U_2$  are *estimated* from finite samples. Perturbations with larger effect sizes produce higher-SNR data, yielding better-estimated subspaces. This estimation quality difference propagates into the distance metric, creating an indirect magnitude sensitivity that is absent from the population-level definition but present in every finite-sample application.

### 3.4 Complementary metrics

In addition to direction instability, we computed three complementary metrics for each LINCS drug: magnitude coefficient of variation ( $\text{CV} = \text{SD}(\|\mathbf{s}_i\|) / \text{mean}(\|\mathbf{s}_i\|)$ ), Fréchet variance (trace of the signature covariance matrix), and mean top-gene Jaccard similarity (overlap of the top 50 most perturbed genes across cell-line pairs). These serve as comparison baselines for the multi-metric benchmark (Section 7.3).

## 4 Part I: The Magnitude Confound

### 4.1 Pre-registration integrity chain

All pre-registrations for the magnitude confound experiments were committed before the author reviewed experimental output (commit 249abaf). The DepMap replication hypothesis was committed separately (commit 98897ed) before execution. The analysis script was frozen (commit 92e3276) before running. The magnitude-confound analysis (Section 4.4) is exploratory and labeled as such.

### 4.2 Experiment: biological corrections do not rerank direction instability

**Hypothesis:** Removing confounding features (toxicity genes in LINCS, essential-gene subspace in Perturb-seq, cell-health features in JUMP-CP) would substantially reorganize the direction instability ranking ( $\rho < 0.70$ ).

**Decision criteria:**  $\rho < 0.70$ : correction bites.  $0.70 \leq \rho < 0.83$ : marginal.  $\rho \geq 0.83$ : does not bite.

**Result:** Corrections do not bite in any domain (Table 1).

Table 1: Rank preservation after biological corrections across three perturbation domains. All Spearman correlations exceed the pre-registered “doesn’t bite” threshold of  $\rho \geq 0.83$ .

| Domain                | Correction                      | $\rho$    | $N$     |
|-----------------------|---------------------------------|-----------|---------|
| LINCS (global)        | Toxicity genes removed          | $> 0.99$  | 8,949   |
| LINCS (HDAC gradient) | Toxicity genes removed          | 0.83      | 20      |
| Perturb-seq           | Essential-gene subspace removed | 0.91      | 1,676   |
| JUMP-CP               | Cell-health features removed    | 0.99      | 25,254  |
| Same-tissue LINCS     | Within-tissue variance genes    | 0.98–0.99 | 176–358 |

This robustness is independently valid because direction instability is cosine-based and magnitude-invariant. The ranking is robust because the underlying geometry is stable, which we now understand is partially because cosine normalization removes the primary source of apparent variation (magnitude).

### 4.3 Experiment: essential-gene divergence replicates in DepMap

**Exploratory prior (disclosed):** In Perturb-seq, essential gene knockdowns showed higher Grassmannian geodesic distance than non-essentials (Cohen’s  $d = 0.55$ , bootstrap CI [0.45, 0.65]).

**Pre-registered prediction:** Essential genes will show higher variability of dependency profiles across cell lines than non-essential genes (one-tailed,  $d > 0.35$ ). Framed as confirmatory replication of the Perturb-seq finding.

**Result:**  $d = 2.65$  (bootstrap CI [2.52, 2.79]). Pre-registered criterion satisfied. PC1 removal changes effect by 0.2%, ruling out the fitness confound.

### 4.4 Discovery of the magnitude artifact

**Status: EXPLORATORY. Not pre-registered. Discovered post-hoc after observing that  $d = 2.65$  was implausibly large.**

The  $d = 2.65$  result triggered scrutiny: Cohen’s  $d$  values above 2 imply near-zero distributional overlap, which is unusual for biological comparisons. Two checks revealed the effect is a magnitude artifact.

#### 4.4.1 DepMap: coefficient-of-variation normalization

Essential genes have large negative Chronos scores (by definition—this is what “essential” means in DepMap). Larger absolute scores mechanically produce larger absolute variance. Normalizing by mean effect magnitude (coefficient of variation:  $SD / |\bar{x}|$ ) removes this confound:

- Essential CV: mean = 0.38, median = 0.26
- Non-essential CV: mean = 2.95, median = 1.71
- Cohen’s  $d$  (CV):  $-1.33$  (sign reversal)

After normalization, non-essential genes show *higher* relative variability. This reversal is biologically interpretable: non-essential genes are dispensable under most conditions but occasionally recruited for context-specific functions, producing high relative variability around a near-zero mean. The raw  $d = 2.65$  was the tautology that “things with bigger effects have bigger variance,” masking a real (and opposite-signed) pattern of context-dependent non-essential gene recruitment.

#### 4.4.2 Perturb-seq: magnitude-matched comparison

Geodesic distance correlates with CERES effect magnitude (Spearman  $\rho = 0.33$ ). After matching essential and non-essential genes by response magnitude ( $n = 127$  matched pairs within 0.2 CERES units):

- Essential mean distance: 2.659
- Non-essential mean distance: 2.671
- Cohen’s  $d$ :  $-0.06$  ( $p = 0.62$ , Mann-Whitney)

The effect vanishes entirely. The original  $d = 0.55$  was driven by essential genes having larger transcriptomic effects, which improves PCA subspace estimation quality and shifts geodesic distances independent of true directional diversity.

#### 4.4.3 Confirming the confounder with a magnitude-invariant metric

We computed PC1 direction instability—the cosine distance between unit-normalized primary response directions in K562 and RPE1—for each gene. Raw PC1 direction instability shows a small essential/non-essential difference ( $d = 0.19$ ,  $p = 0.001$ , magnitude correlation  $\rho = 0.10$ ). After magnitude matching ( $n = 127$  pairs):

- Essential PC1 D (matched): 0.505
- Non-essential PC1 D (matched): 0.512
- Cohen’s  $d$ :  $-0.05$  ( $p = 0.63$ )

After residualization (all 1,567 genes):  $d = 0.07$ ,  $p = 0.19$ .

The magnitude-invariant metric converges to the same null as the magnitude-sensitive one. This rules out the mediator interpretation: if essential genes had genuinely more diverse response *directions* (independent of response strength), the cosine-based metric would detect this regardless of magnitude matching. It does not. The confound operates through estimation quality at every level.

#### 4.4.4 Overriding the pre-registered confirmation

The pre-registered criterion ( $d > 0.35$ ) was technically satisfied by the raw SD metric ( $d = 2.65$ ). We override this on honesty grounds: the metric captured magnitude, not the directional divergence it was intended to measure. A pre-registration protects against cherry-picking hypotheses; it does not oblige one to report a confounded metric as if it measures the intended construct. Both the “confirmation” and the artifact are reported; the latter is our actual conclusion.

### 4.5 Cross-cell-type transport distances

**Status: EXPLORATORY. Discovered post-hoc after running pre-registered transport tests on Perturb-seq data.**

Two pre-registered tests asked whether CRISPR knockdown responses for the *same gene* produce more similar cross-cell-type distances than knockdowns of *different genes*, testing “gene-specific transport” between K562 and RPE1 [Replogle et al., 2022]. The first used geodesic distance on Gr(5, 2073) between top-5 PCA subspaces (1,676 genes); the second used earth mover’s distance (Wasserstein-1) between full perturbed-cell distributions (1,875 genes, subsampled to 200 cells, 10 repeats, exact POT solver).

Both naive results showed same-gene distances lower than random-gene distances: geodesic  $d = -0.14$  ( $p = 2.6 \times 10^{-6}$ ), EMD  $d = -0.172$  ( $p = 9.8 \times 10^{-19}$ ). Both fell in the pre-registered “ambiguous” range ( $|d| < 0.2$ ).

Both metrics correlate strongly with minimum cell count per gene (geodesic: Spearman  $\rho = -0.84$ ; EMD:  $\rho = -0.47$ ). Stratifying by sample size reveals the same bin-inversion pattern as the DepMap artifact:

Table 2: Cross-cell-type distance effect size inverts across sample-size bins for both metrics. Each cell reports Cohen’s  $d$  of same-gene distance relative to the random-gene null.

| min( $n$ ) bin | $n_{\text{geo}}$ | $n_{\text{EMD}}$ | Geodesic $d$ | EMD $d$ |
|----------------|------------------|------------------|--------------|---------|
| [20, 40)       | 204              | 403              | +1.06        | +0.44   |
| [40, 60)       | 407              | 407              | +0.58        | +0.02   |
| [60, 100)      | 575              | 575              | −0.14        | −0.27   |
| [100, 200)     | 429              | 429              | −1.13        | −0.71   |
| [200, 700)     | 61               | 61               | −2.02        | −0.75   |

The monotonic inversion is the diagnostic: if gene-specific transport were biological, all bins would show  $d < 0$  regardless of cell count. Instead, both metrics transition smoothly from positive to negative as  $n$  increases. PCA subspace estimation is more  $n$ -sensitive than distributional EMD, so the metric one would naively trust more (subspace geodesic) is more contaminated.

## 5 Causal DAG

The magnitude confound has the same causal structure across all affected metrics, though the mechanism differs.

### 5.1 DepMap (variance pathway)

$$\text{Magnitude} \rightarrow \begin{cases} \text{Essentiality label} & (\text{confounds grouping}) \\ \text{Score variance} & (\text{inflates metric}) \end{cases}$$

Essentiality is *operationally defined* by dependency-score magnitude ( $\text{Chronos} < -0.5$ ). The magnitude–essentiality association is near-tautological. Magnitude also causes higher variance (larger absolute numbers span larger ranges). Adjusting for magnitude is correct because it removes a spurious path, not a mediating one.

### 5.2 Perturb-seq (estimation pathway)

$$\text{Magnitude} \rightarrow \begin{cases} \text{Essentiality status} & (\text{confounds grouping}) \\ \text{Subspace estimation quality} \rightarrow \text{Geodesic distance} & (\text{inflates metric}) \end{cases}$$

The evidence that this is a confounder rather than a mediator: if magnitude were a *mediator*, then conditioning on magnitude would attenuate but not eliminate the essential/non-essential difference. Instead, magnitude-matching eliminates it completely across all tested metrics: geodesic distance (matched  $d = -0.06$ ), PC1 cosine D (matched  $d = -0.05$ ), and PC1 cosine D residualized over all genes ( $d = 0.07$ ,  $p = 0.19$ ).

### 5.3 Cross-cell-type transport (estimation pathway)

$$\text{Cell count} \rightarrow \begin{cases} \text{Same-gene pairing structure} & (\text{confounds grouping}) \\ \text{Estimation quality} \rightarrow \text{Distance metric} & (\text{deflates metric}) \end{cases}$$

Cell count determines estimation quality for both subspace-based (geodesic,  $\rho = -0.84$ ) and distribution-based (EMD,  $\rho = -0.47$ ) metrics. Same-gene pairs preserve the within-gene cell count correlation between K562 and RPE1, while random-gene pairs break this matching.

### 5.4 LINCS direction instability (no confound)

$$\text{Magnitude} \not\rightarrow D \quad (\text{cosine normalization blocks the path})$$

Direction instability divides by  $\|\mathbf{s}_i\|$  before computing angles. The mechanical path from magnitude to metric is blocked. In finite samples, a residual empirical correlation persists ( $\rho \approx -0.41$  between  $D$  and signal  $\ell_2$  norm in LINCS): weaker signals produce noisier unit vectors, slightly inflating  $D$ . This second-order effect is documented but small relative to the primary confound in subspace metrics ( $\rho = 0.33$ ).

## 6 Part II: Direction Instability Captures Real Biology

With magnitude confounds eliminated from contention, we present evidence that direction instability captures genuine biological signal: it predicts mechanism transport, stratifies by mechanism of action, and reveals fine-grained mechanistic gradients.

## 6.1 The direction instability landscape

Across 8,949 LINCS drugs tested in  $\geq 3$  cell lines, direction instability spans 0 (identical signatures in every context) to 1.33 (near-orthogonal signatures). The distribution is unimodal with a long right tail. Drugs with  $D < 0.2$  produce consistent transcriptomic effects regardless of cellular background; those with  $D > 0.8$  produce context-dependent, divergent responses.

## 6.2 Gene-level validation: Jaccard overlap confirms directional signal

For drugs with low  $D$  (consistent directions), the top 50 most perturbed genes should overlap substantially across cell lines. For drugs with high  $D$  (divergent directions), the perturbed gene sets should differ. Mean top-gene Jaccard similarity correlates with direction instability at  $\rho = -0.72$ , confirming that the geometric metric captures a biological signal: directionally stable drugs perturb the same genes regardless of context.

## 6.3 MOA stratification in LINCS

Drugs targeting fundamental cellular machinery—HDAC inhibitors, proteasome inhibitors, mTOR inhibitors, topoisomerase inhibitors—exhibit low direction instability (mean  $D = 0.28$ ,  $n = 1,127$ ). These targets are conserved across cell types, so the transcriptomic response to their inhibition is consistent. Receptor agonists, GPCR modulators, and ion channel drugs exhibit high direction instability (mean  $D = 0.71$ ,  $n = 463$ ). These targets are differentially expressed across cell types, so the downstream transcriptomic response depends on which receptors are present. The separation is large (Cohen’s  $d = 1.94$ , 95% CI [1.61, 2.33]).

## 6.4 HDAC inhibitor selectivity gradient

Among HDAC inhibitors, direction instability reveals a selectivity gradient invisible to other metrics. Pan-HDAC inhibitors (vorinostat, panobinostat, belinostat) have the lowest  $D$  values ( $D \approx 0.15$ – $0.20$ ): they inhibit all HDAC isoforms and produce near-identical transcriptomic effects in every cell line. Class-selective inhibitors (entinostat, tubacin) have intermediate  $D$  ( $D \approx 0.25$ – $0.35$ ): their effects depend partly on which HDAC isoforms each cell line expresses. Isoform-selective inhibitors targeting context-dependent HDACs (PCI-34051 targeting HDAC8 [Balasubramanian et al., 2008]) have the highest  $D$  values among HDAC drugs ( $D \approx 0.45$ ).

This gradient is predicted by known biology [Bolden et al., 2006]: pan-inhibitors hit all targets regardless of expression context, while selective inhibitors depend on context-specific target availability. The correlation between target breadth and directional consistency confirms that direction instability captures a mechanistically interpretable signal.

## 6.5 Imatinib case study

Imatinib [Deininger et al., 2005] has low direction instability ( $D = 0.18$ , 50th percentile) and high cross-cell-line Jaccard overlap (0.42, top 5%). Its primary target, BCR-ABL, drives a stereotyped transcriptomic response: the same downstream genes are perturbed in every cell line that expresses the target. Cell lines lacking BCR-ABL show minimal response (near-zero signatures), which cosine DI handles correctly—near-zero vectors are treated as uninformative rather than “different.”

## 6.6 Predictive validation: leave-one-out AUROC

If direction instability captures mechanism transport, low- $D$  drugs should produce consistent signatures even in cell lines excluded from the original calculation. We test this with leave-one-out cross-validation: for each drug with  $\geq 10$  cell lines ( $n = 2,694$ ), we compute  $D$  from  $K - 1$  cell lines, then ask whether the held-out signature is consistent (cosine similarity  $> 0.5$  with the remaining signatures).

Direction instability predicts the held-out consistency with AUROC = 0.986 (95% CI [0.983, 0.989],  $n = 2,694$ ). For each drug, the held-out cell line is excluded from both the  $D$  computation and the consistency evaluation, so the predictor and outcome share no cell-line data. A permutation null (shuffling drug labels) yields AUROC =  $0.500 \pm 0.012$  ( $n = 1,000$  permutations), confirming the signal is drug-specific. This cosine-matched AUROC is an exploratory result; the pre-registered construction-neutral benchmark (Section 7.3) reports AUROC = 0.945 [0.937, 0.952] on a Jaccard outcome that cannot benefit from cosine-cosine matching.

## 6.7 Robustness controls

**Replicate noise:** Computing  $D$  from biological replicates within the same cell line yields near-zero values ( $D < 0.05$ ), confirming that technical noise does not inflate the metric.

**Cytotoxicity:** Cytotoxic drugs might produce consistent signatures (low  $D$ ) by killing cells uniformly, yielding a trivial “death program” rather than mechanism-specific transport. Three controls address this. First, restricting to sub-lethal concentrations preserves the AUROC (0.979 vs. 0.986). Second, removing known cytotoxicity genes from the 978-landmark set does not change rankings ( $\rho > 0.99$ , Table 1). Third, known cytotoxic drugs span the full  $D$  range (mean  $D = 0.42$ ), inconsistent with a universal low- $D$  death signal.

**Population null:** Each drug’s  $D$  was tested against a population null—the expected  $D$  from random drug pairs at the same sample size. The  $p$ -value distribution is uniform for most drugs, with enrichment near zero for low- $D$  drugs and near one for high- $D$  drugs, matching the expected pattern for a real signal.

# 7 Part III: Pre-registered Combined Experiments

## 7.1 Pre-registration integrity chain

We pre-registered ten decision criteria across three experiments (commit 8d74bd0), specifying quantitative thresholds and interpretation guides before observing any results. Five criteria passed and five failed. We report all outcomes against the frozen thresholds, interpret the failures, and distinguish pre-registered conclusions from post-hoc observations.

## 7.2 Experiment 1: cosine direction instability on Perturb-seq

This experiment tests the central claim of the magnitude confound analysis: cosine-based direction instability should show flat bins across cell-count strata, unlike geodesic distance and EMD, which invert.

Direction instability detected a small, consistent same-gene transport signal ( $d = -0.22$ , 95% CI  $[-0.33, -0.19]$ ; same-gene mean  $D = 0.152$ , random-gene mean  $D = 0.161$ ). Both C1 and C2 failed their pre-registered thresholds: the aggregate  $|d|$  exceeded 0.1, and per-bin effect sizes fell outside the  $\pm 0.15$  band. The pre-registered scientific hypothesis, however, was immunity to bin inversion—the pattern that invalidates confounded metrics. Across five cell-count bins spanning

Table 3: Pre-registered criteria outcomes. All thresholds were frozen before experiments were run (commit 8d74bd0). “Pass” and “Fail” refer strictly to the pre-registered threshold; interpretation follows in the text.

| Exp. | Criterion                | Threshold                     | Observed                  | Result |
|------|--------------------------|-------------------------------|---------------------------|--------|
| 1    | C1: Aggregate $ d $      | $< 0.1$                       | $d = -0.22$               | Fail   |
| 1    | C2: Bin-flatness         | All bins $\in [-0.15, +0.15]$ | $-0.19$ to $-0.23$        | Fail   |
| 1    | C3: Cell-count $ \rho $  | $< 0.20$                      | $\rho = -0.16$            | Pass   |
| 2    | C1: DI cosine gap        | $\geq 0.02$ over alternatives | $-0.007$                  | Fail   |
| 2    | C2: DI Jaccard gap       | $> 0$ over alternatives       | $+0.088$                  | Pass   |
| 2    | C3: Gap shrinks          | Cosine gap $>$ Jaccard gap    | Reversed                  | Fail   |
| 2    | C4: Correction converges | Corrected gap $<$ raw gap     | $0.088 \rightarrow 0.023$ | Pass   |
| 3    | C1: Group $d > 0.3$      | Machinery $<$ receptor        | $d = 0.77$                | Pass   |
| 3    | C2: Three-tier ordering  | Mach. $<$ kinase $<$ receptor | Kinase $<$ mach.          | Fail   |
| 3    | C3: Significance         | $p < 0.05$                    | $p = 0.002$               | Pass   |

[20, 700) cells, per-bin effect sizes ranged from  $-0.19$  to  $-0.23$ : a total span of 0.04. For comparison, geodesic distance spans 3.08 across the same bins ( $d = +1.06$  to  $d = -2.02$ ); earth mover’s distance spans 1.19 ( $+0.44$  to  $-0.75$ ). The cosine metric shows a consistent shift rather than inversion, and the cell-count correlation is small ( $\rho = -0.16$ , C3 passed). The  $\pm 0.15$  auxiliary band did not anticipate a genuine weak same-gene signal at this magnitude, but the absence of inversion—the primary diagnostic—is confirmed. We report the absolute means and effect size rather than the  $p$ -value ( $2.8 \times 10^{-13}$ ), since a significant-but-small effect at  $n = 1,676$  is precisely the pattern this paper warns against overinterpreting.

### 7.3 Experiment 2: multi-metric LOO benchmark on LINCS

This experiment evaluates direction instability against three alternative metrics (Fréchet variance, magnitude CV, mean top-gene Jaccard) on the predictive benchmark from Section 6.6, using a construction-neutral outcome (Jaccard overlap) to avoid the structural advantage that cosine–cosine matching would confer.

Direction instability achieved the highest AUROC on the construction-neutral Jaccard outcome (0.945, 95% CI [0.937, 0.952] vs. 0.857 [0.841, 0.872] for Fréchet variance; paired  $\Delta = +0.088$  [ $+0.075, +0.102$ ],  $p < 10^{-4}$ ; C2 passed). After magnitude residualization, DI retained its advantage (corrected AUROC 0.754 [0.735, 0.773] vs. 0.731 [0.712, 0.751]; C4 passed). On the cosine outcome, Fréchet variance (0.976 [0.968, 0.983]) exceeded DI (0.969 [0.958, 0.979]; paired  $\Delta = -0.007$  [ $-0.012, -0.002$ ],  $p = 0.002$ ; C1 failed).

C3 failed because DI’s advantage over alternatives was *larger* on the Jaccard outcome than on cosine—the opposite of our prediction that the construction-matched advantage would shrink. This reversal is consistent with the paper’s central thesis: the cosine outcome is defined by the same geometry the matched metric optimizes, so construction matching flatters the matched metric. DI’s advantage emerges on the construction-neutral outcome, where it cannot benefit from geometric self-similarity. The two failures strengthen rather than weaken the central claim, but we report them as failures of the pre-registered predictions.

## 7.4 Experiment 3: MOA class stratification in JUMP Cell Painting

This experiment tests whether the LINCS MOA stratification pattern (Section 5.3) replicates in a different measurement modality: Cell Painting morphological profiles.

Drugs targeting fundamental cellular machinery showed lower direction instability in Cell Painting morphological profiles (mean  $D = 0.69$ ,  $n = 18$ ) than receptor-mediated drugs (mean  $D = 0.83$ ,  $n = 88$ ;  $d = 0.77$ , 95% CI [0.33, 1.35], Mann–Whitney  $p = 0.002$ ; C1 and C3 passed). This replicates the LINCS L1000 pattern (LINCS  $d = 1.94$ ) in a different measurement modality, with expected attenuation.

The three-tier ordering failed (C2): kinase inhibitors (mean  $D = 0.65$ ,  $n = 21$ ) fell slightly below machinery rather than between machinery and receptor. At  $n = 18$  and  $n = 21$ , the 0.04 mean difference between machinery and kinase is well within sampling noise; the distinction between these two groups is underpowered. The robust result is the machinery-vs-receptor separation, which holds with a large effect size across measurement technologies.

## 8 The Fix

Three normalizations remove the magnitude confound from affected metrics:

1. **Coefficient of variation** (SD / |mean effect|): removes scale dependence. Applicable when effects have non-zero means.
2. **Magnitude matching**: compare only perturbations with similar absolute effect sizes. The most assumption-free approach; does not require linearity.
3. **Residualization**: regress out effect magnitude and test residuals. Assumes a linear (or specified) relationship between magnitude and the metric.

Cosine-based formulations (direction instability, Equation 1) require no correction—they are the magnitude-normalized version by construction.

## 9 Discussion

This paper makes two claims. First, variance-based, subspace-based, and distribution-based perturbation-divergence metrics are magnitude-confounded when used to compare groups that differ in effect strength or sample size. Second, cosine-based direction instability is immune to this confound and captures genuine biological signal: mechanism transport, MOA stratification, and mechanistic selectivity gradients.

The magnitude confound claim is supported by three instances of the same discovery pattern: a pre-registered metric yields an apparently meaningful effect (DepMap  $d = 2.65$ , Perturb-seq geodesic  $d = 0.55$ , cross-cell-type transport  $d = -0.14$  to  $-0.172$ ), which post-hoc analysis reveals as estimation-quality confounding. In each case, the signal inverts or vanishes after controlling for the confound.

The direction instability claim is supported by predictive validation (LOO AUROC = 0.945 [0.937, 0.952] on the construction-neutral outcome; 0.986 [0.983, 0.989] on the exploratory cosine-matched outcome), cross-technology MOA replication (LINCS  $d = 1.94$ , JUMP-CP  $d = 0.77$ ), and pre-registered experiments where five of ten criteria passed. The criteria that failed are interpretable: the Perturb-seq bin-flatness band did not anticipate a genuine weak transport signal (bins span 0.04, vs. 3.08 for geodesic—no inversion, but shifted outside the  $\pm 0.15$  band), Fréchet

variance exceeded DI on the cosine outcome by 0.007 AUROC (paired  $p = 0.002$ ), and the three-tier MOA ordering was underpowered at  $n = 18$ –21.

## 9.1 Pre-registration as diagnostic

Half of our pre-registered thresholds missed. We treat this as informative rather than embarrassing. The failures reveal where our assumptions were wrong: the Perturb-seq bin-flatness threshold assumed the cosine metric would show near-zero aggregate effect, when in fact a genuine weak signal exists; the cosine–Jaccard gap prediction assumed DI’s advantage would be construction-dependent, when it turned out to be construction-independent; Fréchet variance’s marginal win on the cosine outcome ( $\Delta = 0.007$ ) is consistent with the construction-matching critique the paper advances. Each failed prediction is itself a finding.

The practice of overriding a technically-passed criterion (DepMap  $d = 2.65$ ) on honesty grounds deserves emphasis. A pre-registration protects against cherry-picking hypotheses; it does not oblige one to report a confounded metric as if it measures the intended construct. Rigid adherence to pre-registered decisions when the decision criteria themselves are revealed as inadequate serves pre-registration theater, not scientific integrity.

## 9.2 Implications for perturbation biology

The practical implication is a rule: before comparing “divergence” across groups that differ in perturbation strength or cell count, normalize for magnitude. Failing to do so produces effects that look large ( $d = 2.65$ ,  $d = 0.55$ ) or directionally misleading ( $d = -0.14$  averaging over bins from  $+1.06$  to  $-1.24$ ) but reduce to the tautology that bigger effects are noisier or better-estimated.

Researchers computing variability metrics on perturbation atlases (DepMap, Perturb-seq, PRISM, JUMP-CP) should verify that their metric is magnitude-invariant before interpreting group differences as biological divergence. The transport confound is particularly insidious because it can masquerade as a weak positive signal: the aggregate  $d = -0.14$  appeared to be a real-but-small biological effect, when it was an average over bins pointing in opposite directions.

Direction instability offers a constructive alternative. Its magnitude immunity is mechanical (cosine normalization blocks the path from magnitude to metric), and its biological validity is confirmed by predictive performance (construction-neutral LOO AUROC = 0.945 [0.937, 0.952]), cross-technology replication, and mechanistically interpretable gradients within drug classes.

## 9.3 Limitations

The magnitude-matching analysis used  $n = 127$  matched pairs, constrained by the overlap in effect-size distributions between essential and non-essential genes. The matched sample represents the narrow region where both distributions overlap: atypically weak essential genes matched to atypically strong non-essential genes. However, the residualization analysis uses all 1,567 labeled genes and converges to the same null ( $d = 0.07$ ,  $p = 0.19$  for PC1 D), confirming that the matching result is not an artifact of sample restriction.

Our claim that direction instability is “immune” to magnitude confounding is precise: the cosine normalization removes the *mechanical* dependency. A residual empirical correlation persists in finite samples ( $\rho = 0.10$  for PC1 D vs. magnitude in Perturb-seq;  $\rho \approx -0.41$  between  $D$  and signal norm in LINCS drug data). This arises because weaker signals produce noisier estimated directions, slightly inflating  $D$ . The metric is mechanically immune but not entirely empirically independent of magnitude when directions are estimated from finite data.

The JUMP Cell Painting MOA stratification ( $n = 18$  machinery,  $n = 21$  kinase,  $n = 88$  receptor) is underpowered for fine-grained ordering. The machinery-vs-receptor separation is robust ( $d = 0.77$ ,  $[0.33, 1.35]$ ,  $p = 0.002$ ), but the kinase-machinery distinction ( $\Delta = 0.04$ ) is not resolvable at these sample sizes.

The Perturb-seq analysis uses only two cell types (K562, RPE1), so direction instability reduces to a single pairwise cosine rather than a population average. This suffices for the bin-flatness test (the diagnostic question is whether per-bin  $d$  values invert with cell count, not whether the absolute level of transport is high) but limits generalization to multi-context settings.

## 10 Conclusion

We pre-registered divergence and transport predictions, naive metrics confirmed them, and our own post-hoc checks killed them—three times with the same pattern. The killing is the finding: geometric perturbation metrics measure magnitude and sample size unless explicitly prevented from doing so. The DAG is simple, the fix is simple, and the failure mode is likely widespread in any analysis comparing perturbation groups that differ in effect strength or cell count.

The constructive counterpart: direction instability, which blocks the magnitude path by construction, predicts cross-cell-line mechanism transport (construction-neutral LOO AUROC = 0.945  $[0.937, 0.952]$ ), replicates MOA stratification across measurement technologies, and resolves mechanistic gradients within drug classes. The metric works because it asks the right question—do perturbation effects point in the same direction?—rather than a question contaminated by how precisely we estimated them.

## Data and Code Availability

All analysis code and pre-registration documents are available at <https://github.com/elliotttower/bracket-norm-validity> and <https://github.com/elliotttower/drug-perturbation-geometry>. The repositories contain the full commit history demonstrating the pre-registration integrity chains (commits 1dc20a2, 249abaf, 98897ed, 92e3276, 8d74bd0). Zenodo DOIs: drug transport dataset ([zenodo.org/records/21207728](https://zenodo.org/records/21207728)), magnitude confound replication materials ([zenodo.org/records/21207429](https://zenodo.org/records/21207429)). DepMap data (24Q4 release) is available from <https://depmap.org/portal/>. LINCS L1000 data is from GEO (GSE92742). Perturb-seq data is from Replogle et al. [2022]. JUMP Cell Painting data is from Chandrasekaran et al. [2023].

## Author Contributions

E.T. conceived the study, designed experiments, wrote pre-registrations, implemented all analyses, discovered the magnitude artifact, and wrote the manuscript.

## AI Disclosure

Claude Code (Anthropic) was used as a coding assistant for data analysis, statistical computation, and manuscript preparation. Perplexity Pro was used for literature search and as an independent reviewer of pre-registration documents and experimental scripts. All scientific decisions, experimental design, interpretations, and conclusions are the author’s.

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## A MOA Classification Statistics

Table 4: MOA class sizes used in stratification analyses. LINCS classification uses Drug Repurposing Hub annotations. JUMP-CP classification uses the same MOA lists applied to InChIKey-matched compounds.

| MOA class                                              | LINCS $n$ | JUMP-CP $n$ |
|--------------------------------------------------------|-----------|-------------|
| Machinery (HDAC, proteasome, mTOR, topoisomerase, ...) | 1,127     | 18          |
| Kinase inhibitors                                      | 892       | 21          |
| Receptor (GPCR, ion channel, nuclear receptor, ...)    | 463       | 88          |
| Other / unclassified                                   | 6,467     | 203         |

Table 5: Direction instability of HDAC inhibitors in LINCS L1000, ranked by  $D$ . Pan-HDAC inhibitors cluster at low  $D$ ; isoform-selective inhibitors at high  $D$ .

| Drug         | Selectivity | $D$  | $n_{\text{cells}}$ | Clinical phase |
|--------------|-------------|------|--------------------|----------------|
| Panobinostat | Pan-HDAC    | 0.14 | 64                 | Approved       |
| Vorinostat   | Pan-HDAC    | 0.16 | 63                 | Approved       |
| Belinostat   | Pan-HDAC    | 0.19 | 42                 | Approved       |
| Entinostat   | Class I     | 0.27 | 38                 | Phase III      |
| Tubacin      | HDAC6       | 0.32 | 25                 | Tool compound  |
| PCI-34051    | HDAC8       | 0.45 | 18                 | Preclinical    |

## B HDAC Inhibitor Selectivity Gradient

## C Top 50 Directionally Stable Drugs

The 50 drugs with lowest direction instability among those tested in  $\geq 10$  cell lines are dominated by: pan-HDAC inhibitors, proteasome inhibitors (bortezomib, carfilzomib), HSP90 inhibitors (tanespimycin, alvespimycin), and topoisomerase inhibitors (irinotecan, topotecan). All are cytotoxic agents targeting highly conserved cellular machinery, consistent with the mechanism-transport interpretation. The full ranked list with  $D$  values,  $n$  cell lines, MOA, and targets is provided in the Zenodo supplement.